

MD. MOSFIKUR RAHMAN

Software Engineer | Applied ML & Backend Systems

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Google Scholar](#)

SUMMARY

Software engineer with 4+ years of production backend experience (microservices, rule engines, distributed data pipelines) and a research record of 10 peer-reviewed publications in applied machine learning and data analysis. Comfortable across the full path from messy real-world data to systems that run at scale.

EDUCATION

Daffodil International University

B.Sc. in Computer Science and Engineering — CGPA: 3.83/4.00

Dhaka, Bangladesh

2018 – 2021

Adam Mickiewicz University

Erasmus+ International Exchange (one semester)

Poznań, Poland

2021

EXPERIENCE (4+ YEARS)

Akij iBOS Ltd.

Software Engineer L-II

Dhaka, Bangladesh

Nov. 2024 – Present

- Own end-to-end architecture for the Online Travel Agency (OTA) domain — supplier configuration, pricing rules, and booking workflows.
- Designed a Drools-based rule engine for context-aware pricing and decision-making.
- Led backend design for supplier orchestration and aggregated response generation in a microservices architecture (Java, .NET, Node.js).
- Deployed services with Docker and Kubernetes; used gRPC for inter-service communication and Azure/AWS in production.

REVE Systems Ltd.

Software Engineer

Dhaka, Bangladesh

Jul. 2023 – Oct. 2024

- Built backend systems in Java, Spring Boot, and Oracle for the Customs Bond Management System, a National Board of Revenue (NBR) platform.
- Improved system throughput by ~25% through query tuning and data-access optimization.
- Delivered the Legal Case and Utilization Declaration modules; built BGMEA/BKMEA database sync for export compliance.
- Implemented OAuth2 / Spring Security authentication and clean-architecture REST APIs.

BJIT Group

Software Engineer (Backend — Java & GraphQL)

Dhaka, Bangladesh

Apr. 2022 – Jun. 2023

- Built GraphQL Backend-for-Frontend services for Rakuten e-commerce modules using Java and Spring Boot.
- Implemented resolver batching and schema stitching, reducing nested-query latency.
- Optimized database access paths and contributed reusable backend service patterns and internal tooling.

SELECTED PROJECTS

OTA Backend System

Akij iBOS Ltd.

- Microservices-based travel platform for booking, pricing, and supplier integration with a Drools pricing engine.
- Designed service boundaries and domain models for scalable, maintainable architecture.

Customs Bond Management System (NBR)

REVE Systems Ltd.

- Legal Case and Utilization Declaration modules; BGMEA/BKMEA sync for garment-export compliance.

Rakuten Echiba Backend

BJIT Group

- GraphQL BFF for e-commerce modules; reduced query latency via resolver and join-path optimization.

TECHNICAL SKILLS

Languages: Java, C#, JavaScript, SQL, Python

Backend: Spring Boot, Spring Security, JPA/JDBC, GraphQL, .NET, Node.js

Architecture: Microservices, REST APIs, Distributed Systems, Modular Monolith

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD

Databases: Oracle, PostgreSQL, SQL Server, MySQL

Technologies: OAuth2, JWT, gRPC, Camunda, Drools

Concepts: OOP, Design Patterns, Clean Architecture, Query Optimization, Secure APIs

Tools: Git, Linux/Unix

RESEARCH & PUBLICATIONS

10 peer-reviewed publications (3 journal, 7 conference). † = first author.

JOURNALS

- † Impact of COVID-19 on mental health: A quantitative analysis of anxiety and depression based on regular life and internet use
Current Research in Behavioral Sciences, 2021, 2, 100037
- COVID-19 and Bangladesh: Situation report, comparative analysis, and case study
Current Research in Behavioral Sciences, 2021, 2, 100034
- Impactful e-learning framework: A new hybrid form of education
Current Research in Behavioral Sciences, 2021, 2, 100038

CONFERENCES

- † Future City of Bangladesh: IoT Based Autonomous Smart Sewerage and Hazard Condition Sharing System
IEEE WIECON-ECE, 2020
- † TraFoo: An Android Application for Food Delivery in Train
IEEE SMART GENCON, 2021
- Cyber Security Intruder Detection Using Deep Learning Approach
Springer: Information Systems & Management Science, 2023
- A New Chaotic-Based Analysis of Data Encryption and Decryption
Springer: Advances in Data Science and AI, 2023
- Machine Learning-Based Prediction of COVID-19: A Robust Approach for Early Diagnosis and Treatment
Springer: Trends in Computational and Cognitive Engineering, 2024

EXTRA-CURRICULAR ACTIVITIES

- **Conference Speaker**
Delivered a technical session at the 10th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2024 (IEEE WIECON-ECE 2024).
- **Guest Speaker**
"How to Get Started: Undergraduate Research Journey", Computer & Programming Club, Daffodil International University.
- **Vice President, Research & Career Wing**
Led mentorship programs, research workshops, and career initiatives at the Computer & Programming Club, DIU (2021).
- **Journal Reviewer**
Reviewer for several Elsevier/Springer journals and international conferences in ML and data science.